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A Comparative Study of Major Multimodal Sentiment Analysis Approaches

A Survey of Fusion Paradigms in Text, Audio, and Visual Sentiment Modeling

Abstract

Multimodal Sentiment Analysis (MSA) has emerged as a vital area of research at the intersection of Natural Language Processing (NLP), Computer Vision, and Speech Processing. Unlike unimodal sentiment analysis, which relies solely on textual information, MSA integrates heterogeneous data streams — text, acoustic, and visual — to infer emotional states and sentiments with higher accuracy. This paper presents a comparative study of the major fusion paradigms in MSA. We analyze Early Fusion, Late Fusion, Tensor-Based Fusion, Attention and Cross-Modal Transformers, and Representation Decoupling. Each approach is evaluated based on its operational modalities, fusion strategy, representative model architectures, common benchmark datasets, key advantages, and inherent limitations.

1. Introduction

With the rapid growth of video-sharing platforms and social media, human communication has become inherently multimodal. Deciphering human affect requires understanding not only what is said (textual content), but also how it is said (vocal intonation, acoustic pitch) and what visual cues accompany it (facial expressions, body language).

Multimodal Sentiment Analysis (MSA) aims to harvest these complementary signals to overcome the ambiguity of unimodal systems, such as detecting sarcasm, subtle emotional shifts, or implicit attitudes. The core challenge in MSA lies in multimodal fusion — the methodology by which disparate, asynchronous, and heterogeneous feature streams are combined into a cohesive joint representation.

This survey categorizes and compares the primary fusion paradigms established in MSA literature, providing a structured overview of their strengths, trade-offs, and typical use cases.

2. Comparative Study Table

Table 1 summarizes the five major MSA fusion paradigms — Early Fusion, Late Fusion, Tensor-Based Fusion, Attention/Cross-Modal Transformers, and Representation Decoupling — across modalities, fusion strategy, representative architectures, benchmark datasets, advantages, and limitations.

Table 1. Comparison of Major Multimodal Sentiment Analysis Fusion Paradigms

Approach / Paradigm	Modalities	Fusion Strategy	Models / Architectures	Benchmark Datasets	Key Advantages	Limitations
Early Fusion (Feature-Level)	Text, Audio, Visual	Feature Concatenation: concatenates raw or pre-extracted unimodal feature vectors prior to classification.	SVMs, Random Forests, Feedforward DNNs, Early MLPs	CMU-MOSI, CMU-MOSEI, IEMOCAP	Simple to implement. Learns low-level cross-modal interactions directly from raw data.	Fails to handle temporal/asynchronous alignment across modalities. Highly sensitive to noise and dimensional imbalance between feature vectors.
Late Fusion (Decision-Level)	Text, Audio, Visual	Decision Aggregation: trains independent classifiers per modality and merges predictions via voting, averaging, or meta-learners.	Modality-specific LSTMs/Transformers + Stacking Classifier / Logistic Regression	CMU-MOSI, MELD, UR-FUNNY	Robust to missing or corrupted modalities at inference time. Allows modality-customized architecture selection.	Ignores fine-grained cross-modal feature interactions during extraction. Higher overall model parameter count.
Tensor-Based Fusion (3D Outer Product)	Text, Audio, Visual	Intermediate Tensor Dynamics: computes explicit unimodal, bimodal, and trimodal interactions via multidimensional outer products.	Tensor Fusion Network (TFN), Low-rank Multimodal Fusion (LMF)	CMU-MOSI, CMU-MOSEI, SIMS	Explicitly models all sub-dimensional modal interactions. Does not require prior temporal feature alignment.	High computational complexity. Outer products scale exponentially with modality dimensions (partly mitigated by low-rank approximations).
Attention & Cross-Modal Transformers	Text, Audio, Visual	Intermediate Cross-Attention: uses directional cross-modal attention to project representations from one modality to another.	Multimodal Transformer (MulT), MAG-BERT, MISA	CMU-MOSI, CMU-MOSEI, CH-SIMS, IEMOCAP	Handles end-to-end learning on unaligned sequences. Dynamically captures complex, long-range cross-modal dependencies.	Requires large labeled datasets. Computationally heavy. Vulnerable to domain shift and acoustic/visual noise.
Representation Decoupling (Specific vs. Invariant)	Text, Acoustic, Visual	Factorized Intermediate Fusion: disentangles inputs into modality-invariant (shared) and modality-specific (unique) latent subspaces before fusion.	MISA, Self-MM, Centroid-focused disentangled networks	CMU-MOSI, CMU-MOSEI, SIMSv2	Minimizes redundant modal noise. Captures uniquely expressive information (e.g., tone vs. text) without dominant-modality bias.	Complex loss functions requiring careful tuning (similarity, orthogonality, reconstruction losses). Slower optimization convergence.

3. Key Findings & Discussion

3.1 Temporal Asynchrony & Alignment

Early approaches relied on explicit forced-alignment tools (such as P2FA) to synchronize audio and video frames with word tokens. Modern Attention/Transformer-based architectures (e.g., MulT) bypass explicit alignment by implicitly mapping cross-modal dependencies using dynamic cross-attention layers.

3.2 Modality Dominance

Text consistently serves as the primary predictor in benchmark datasets like CMU-MOSEI and MELD. Visual and acoustic modalities act primarily as intensity modifiers or disambiguators, refining rather than replacing the sentiment signal carried by language.

3.3 Disentangled Representations

Representation Decoupling techniques mitigate the risk of the textual modality overwhelming audio and visual streams by learning distinct invariant (shared) and specific (unique) subspaces prior to final fusion, improving both performance and interpretability.

4. Conclusion

Multimodal Sentiment Analysis has evolved significantly from simple feature concatenation and decision-level voting to sophisticated cross-modal attention transformers and disentangled feature representations. While attention mechanisms and tensor-based approaches offer superior performance in capturing fine-grained inter-modality dynamics, they demand substantial computational resources and clean alignment. Representation decoupling represents a promising frontier for reducing redundancy and improving model interpretability. Future work in MSA must continue to address real-world challenges, including robustness against missing modalities, computational efficiency for edge deployment, and cross-cultural affective generalizability.